

HIMANSHU RAJNISH BORKAR | ECE (COMPUTER ENGINEERING)

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EDUCATION

Program	Institution	%/CGPA	Year
MS, Electrical and Computer Engg.	University of California San Diego	Incoming FA26	2026
B.Tech, Electrical Engg.	Indian Institute of Technology Madras	8.18 /10.0	2026

POSTER PRESENTATION

- **Himanshu Borkar**, Shrey Ramanujan, Sivarama Krishnan, et al.
“Implementation of a compact multipass cell pulse compressor system with a second-order autocorrelator” presented at *12th DAE-BRNS Ultrafast Sciences (UFS) 14th November 2025. (Poster - P086)*

RESEARCH EXPERIENCE

DISCRETE GATE-SIZING BASED GLITCH MANIPULATION FOR HARDWARE SECURITY

JANUARY '26 - PRESENT

(Guide: [Prof. Janakiraman Viraraghavan](#), Electrical Engineering, IIT Madras)

- Developed a **simulated-annealing** based timing/area/power opt. discrete gate-sizing tool for combinational ckts.
- Analyzed discrete gate size-selection scoring systems using greedy stochastic and **Multi-Armed Bandits** frameworks.
- Analyzing the effect of **glitch minimisation and maximisation** on **power side channel** attacks on crypto-hardware
- Implementing timing/area/power opt. on large combination circuits using **graph partitioning**, and block-wise opt.
- Explored **GNN architectures** for area-min gate-sizes solutions and glitch-window prediction on ISCAS-85 benchmarks

MIN-DELAY GEOMETRIC PROGRAMMING & POST TAPE-OUT GLITCH MITIGATION

JULY '25 - DECEMBER '25 (INTEGRATED CIRCUITS AND SYSTEMS GROUP, IIT MADRAS)

(Guide: [Prof. Janakiraman Viraraghavan](#), Electrical Engineering, IIT Madras)

- Implemented **glitch window estimation** of circuit nodes as a function of logic depth using **Monte Carlo simulations**.
- Developed a workflow for glitch-count extraction from **VCD files**, analyzed trends in glitch window for ISCAS-85 ckts.
- Analyzed and debugged the design of **control logic** implemented in a recently taped-out **In-Memory Compute chip**.
- Diagnosed RTL block responsible for haphazard glitches in the control logic, used **DVFS** to minimize these glitches.

NOVEL AUTOCORRELATOR DESIGN FOR HHG APPLICATIONS

AUGUST '24 - APRIL '25 (FEMTOSCIENCE FACILITY, IIT MADRAS)

(Guide: [Prof. Sivarama Krishnan](#), Department of Physics, IIT Madras)

[Github](#)

- Implemented an Autocorrelator design using the **Mach-Zehnder Interferometer** for coherent 800nm Ultrafast laser.
- Estimated the **pulse width** with the autocorrelator design using a **Second Harmonic Generation** (non-linear) crystal.
- Implemented **Multi-Pass cells** for pulse compression using gas-filled Herriott Cell to achieve femtosecond laser pulses.
- The work and findings were submitted and presented as a poster at the conference, **Ultrafast Science 2025, IIT Hyd**.

COURSE PROJECTS

EMBEDDED MEMORY DESIGN - COMPUTE-IN MEMORY ENGINE DESIGN

JANUARY'26 - MAY '26

(Guide: [Prof. Janakiraman Viraraghavan](#), Electrical Engineering, IIT Madras; [Prof. Rahul M Rao](#), IBM)

[Github](#)

- Designed a Charge-Domain based **9T-1C Compute-In Memory engine** to run inferences on the MNIST image dataset.
- Analyzed the dataset to obtain best **quantization** for weights, training data, **ADC resolution** & information deficit.
- Designed and simulated compute column and a **switched-cap adder** for adder-first architecture for opt. bank count.
- Designed a **stepwise charge-discharge** input driver-based DAC for improved energy efficiency. Performed Monte Carlo simulations on effective compute column. The design achieved only **0.64%** accuracy drop from software baseline.

GATE SIZING FOR AREA MINIMIZATION USING GEOMETRIC PROGRAMMING

JANUARY'25 - MAY '25

(Guide: [Prof. Janakiraman Viraraghavan](#); [Prof. Ramprasath S](#), Electrical Engineering, IIT Madras)

[Github](#)

- Developed a **Geometric Programming**-based gate sizing algorithm for Area Minimization in ISCAS-85 circuits.
- Evaluated T_{wall} and **critical path** for the circuits and analyzed **optimal drive strengths** of standard library gates.
- Implemented **Branch & Bound** algorithm to obtain discretized gate-sizing solution with area-minimum objective.

NON VOLATILE MEMORY-BASED COMPUTE IN-MEMORY ENGINE

JULY'25 - NOVEMBER '25

(Guide: [Prof. Bhaswar Chakrabarti](#), Electrical Engineering, IIT Madras)

[Github](#)

- Designed two-layer **RRAM crossbar arrays** to enable efficient in-memory computation for alphabet classification tasks.
- Performed **hardware-aware training** to approx. binarize weights & minimize loss in hardware inference accuracy.
- Designed and simulated RRAM crossbar arrays with **CMOS activation** & MUX circuits to quantify energy consumption.

DESIGN AND STA OF 16-BIT TREE ADDERS

JUL'25 - NOV '25

(Guide: [Prof. Vinita Vasudevan](#); [Prof. Ramprasath S](#), Electrical Engineering, IIT Madras)

[Github](#)

- Designed schematics and layouts for Mirror-symmetric Ripple Carry FA, 8-bit Full Adder and **16-bit tree adders** like Brent Kung, Kogge-Stone & Sklansky Adders with **propagate/generate** modules method using **hierarchical blocks**.
- Analyzed **post-layout parasitics** and STA reports using **OpenLANE**, determined the max operating frequencies.
- Designed a **Carry-bypass Adder**; synthesized it with Yosys for STA report, identifying **critical path** and slack.

MONTÉ CARLO SIMULATIONS OF IONIC SPECIES FOR ENERGY ANALYSIS

JANUARY'25 - MAY '25

(Guide: [Prof. Yasir Iqbal](#), Department of Physics, IIT Madras)

[Github](#)

- Investigated effects of geometric frustration on low-temperature behavior of magnets using ferromagnetic and anti-ferromagnetic **Ising Models** for square and triangular lattices by implementing **Classical Monte Carlo**.
- Implemented **Variational Monte Carlo** with Metropolis-Hastings algorithm methods for investigation of 1-D **Quantum Harmonic Oscillator**, H-atom and analyzed the cusp and no-cusp conditions for He^{+1} , Li^{+2} and Be^{+3} ionic species.

POSITIONS OF RESPONSIBILITY

COORDINATOR - ETHICAL TECH SUMMIT

MAY'23 - JANUARY '24

- Organized the first-ever student-run **International summit** securing 160 paid delegations. Invited 21 speakers & panelists lectures and present workshops on the Ethical Use of technology.
- Networked with professors in the field of **responsible AI** & Ethics, alongside hosting **ex-White House Tech Advisor**.
- Raised awareness for Ethical use of Technology through a wall painting competition at IIT Madras among 6 teams.

COURSE TEACHING ASSISTANT

JULY '25 - NOVEMBER '25

(Instructor: [Prof. Janakiraman Viraraghavan](#), Electrical Engineering, IIT Madras)

- Computer Organization** - Conducted weekly **tutorials** for 50 students, while clarifying concepts for thorough understanding. Designed **application-based demo sessions** to bridge theory with **practical implementation**.

RELEVANT COURSES

Digital IC Design	Embedded Memory Design	Modeling and Optimization of VLSI
Computer Architecture	Computer Organization	Devices for AI & Neuromorphic computing
Digital Systems & Lab	Microprocessor Theory and Lab	Analog Systems
Deep Learning	Methods of Computational Physics	Quantum Computing & Information

SKILLS

- Programming languages & tools:** C, Python, MATLAB, ARM & x86 Assembly, Linux, Git
- Analysis and Simulation:** Cadence Virtuoso, Cadence Genus, LTSpice, xSchem, Klayout, OpenSTA, Synopsys, ADS

VOLUNTEERING AND CO-CURRICULAR ACHIEVEMENTS

AVANTI FELLOWS - MENTOR

- Mentored underprivileged students through their academic and mental preparation for college entrance exams.
- Traveled monthly in groups to schools to guide mentees and strengthen **mentor-mentee bonding**.

TENNIS

- Competed in multiple ranked **All India Tennis Association** (AITA) Tournaments (South India Zone).
- Secured **3rd place** in IIT Madras' Inter-Hostel Tennis (Doubles) competition.